

Fourier Transform

①

→ (TFT) means transform discrete :

To analyze discrete signals in fourier domain, we have two tools.

- (i) Discrete Time Fourier Transform (DTFT)
- (ii) Discrete Fourier Transform (DFT)

DTFT:-

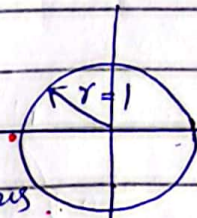
By definition $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$

'n' is transformed into 'ω'

Discrete time $n \rightarrow \omega$ freq.

'n' is discrete $0 \rightarrow 2\pi$ ω is continuous

and periodic always.



IDTFT:-

$$x(n) = \frac{1}{2\pi} \int_0^{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

because 'ω' is continuous

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Discrete Fourier Transform

-: Discrete Fourier Transform (DFT) :-

$$x(n) \xrightleftharpoons[\text{IDFT}]{\text{DFT}} X(k)$$

By defn. DFT :-

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi nk}{N}} \quad \dots \dots \dots k=0, 1, \dots, N-1$$

IDFT:-

$$x(n) = \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi nk}{N}}$$

$$n=0, 1, \dots, N-1$$

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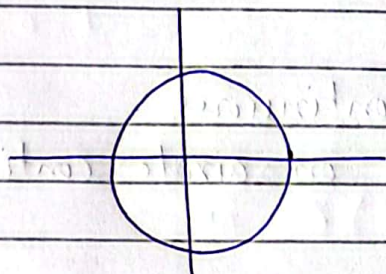
Relation b/w DTFT & DFT

DTFT

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

DFT

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi nk}{N}}$$

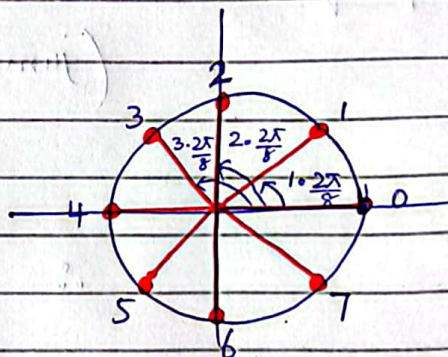


If we look at both equations

$$\omega = \frac{2\pi k}{N}$$

$$\omega = 0 \rightarrow 2\pi$$

Suppose chop off ω into 8 equal parts



$$\text{Total} = 2\pi$$

$$\text{Divide into 8 parts} \rightarrow \frac{2\pi}{8}$$

$$0. \frac{2\pi}{8}$$

$$3. \frac{2\pi}{8}$$

$$1. \frac{2\pi}{8}$$

$$2. \frac{2\pi}{8}$$

$N \rightarrow$ Number of divisions.

$\therefore$ Took $\omega(0 \rightarrow 2\pi)$, Chopped it off into equal number of parts. Now, this ω is

④

Relation b/w DTFT & DFT
Converted into $\frac{2\pi k}{N}$

$$x(n) \xrightarrow{\text{DTFT}} X(\omega) \xrightarrow[\text{Discretize } \omega]{\text{DFT}} X(k)$$

In DTFT

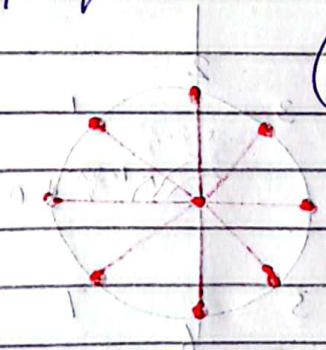
freq. domain is Continuous
(move on circle continuously)

In DFT

freq. domain is discrete

(jump on circle on discrete points)

nothing in between



$$\therefore k = 0, 1, \dots, N-1$$

$\therefore$ DFT is freq. sampled version of

DTFT

For $N=8$

Discrete

also $H(z)$ is sampled, ($x[n] \rightarrow 0$) for $n > N$

then with N steps for continuous time

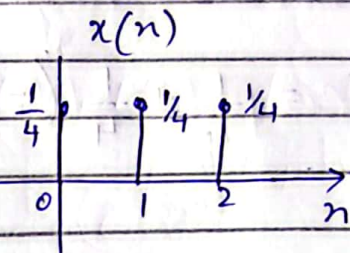
(5)

∴ DFT by Definition :-

Exp

$$x(n) = \begin{cases} 1/4 & 0 \leq n \leq 2 \\ 0 & \text{o.w} \end{cases}$$

$$X(k) = ?$$



Solution

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi nk/N}$$

what is N ??

from question $0 \leq n < 2$

$$N-1 = 2$$

$$N = 3$$

∴ it is 3 pt DFT

$$X(k) = \sum_{n=0}^3 x(n) e^{-j2\pi nk/3}$$

$$X(k) = x(0) e^{-j2\pi(0)k/3} + x(1) e^{-j2\pi(1)k/3} + x(2) e^{-j2\pi(2)k/3}$$

$$X(k) = x(0) + x(1) e^{-j2\pi k/3} + x(2) e^{-j4\pi k/3}$$

$$\text{now } k = 0, 1, \dots, N-1 \\ = 0, 1, 2$$

$$k=0$$

$$X(0) = x(0) + x(1) e^0 + x(2) e^0$$

$$= 1/4 + 1/4 + 1/4 = \boxed{3/4 = X(0)}$$

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put $k=1$ in definition of DFT:

$$X(1) = \frac{1}{4} + \frac{1}{4} e^{-j\frac{2\pi}{3}} + \frac{1}{4} e^{-j\frac{4\pi}{3}}$$

$$= \frac{1}{4} + \frac{1}{4} \left[\cos \frac{2\pi}{3} - j \sin \frac{2\pi}{3} \right] + \frac{1}{4} \left[\cos \frac{4\pi}{3} - j \sin \frac{4\pi}{3} \right]$$

$$= \frac{1}{4} + \frac{1}{4} (-0.5) - j \frac{1}{4} (0.866) + \frac{1}{4} (-0.5) + j \frac{1}{4} (0.866)$$

$$\boxed{X(1) = 0}$$

put $k=2$

$$\boxed{X(2) = 0}$$

$$X(k) = \begin{bmatrix} 3/4 \\ 0 \\ 0 \end{bmatrix}$$

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∴ Problem with Direct Method :-

e.g. $x(n) = \{1, 2, 3, 4\}$ $N=4$

$$X(k) = \sum_{n=0}^{3} x(n) W_N^{nk} \quad \text{--- (1)}$$

$$W_N = e^{-j\frac{2\pi}{N}}$$

(Twiddle factor)

~~$$W_N^{nk} = e^{-j\frac{2\pi}{N}nk}$$~~

$$W_N^{nk} = e^{j\frac{2\pi}{N}nk}$$

$$e^{j\frac{2\pi}{N}nk}$$

$$W_N^{nk}$$

$$W_N^{nk}$$

expand eq (1)

$$X(k) = x(0) W_4^{(1)k} + x(1) W_4^{(2)k} + x(2) W_4^{(3)k} + x(3) W_4^{(4)k}$$

$$X(0)$$

$$X(1)$$

$$X(2)$$

$$X(3)$$

Total Complex multiplications

$$4 \times 4 = 16$$

Total Complex additions

$$4 \times 3 = 12$$

... bottom level direct address :-

Now if $N = 256$.

Complex multiplications $= 256 \times 256$

It means as 'N' increases, calculations become huge and time required to compute DFT will also increase.

SOLUTION:-

An algorithm was proposed to

perform fast computation of Fourier

Transform. This is called

Fast Fourier Transform (FFT)

(Divide and rule).

So divide the sequence.

i.e. N -pt DFT

$N/2$ pt

$N/2$ pt

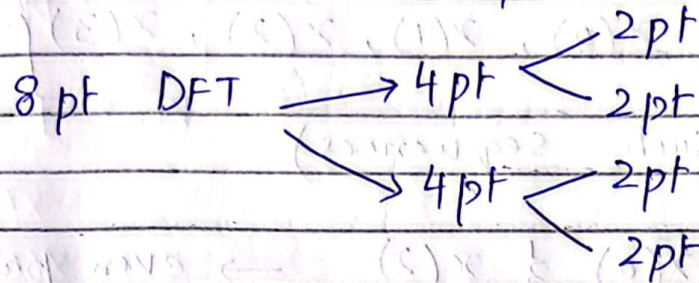
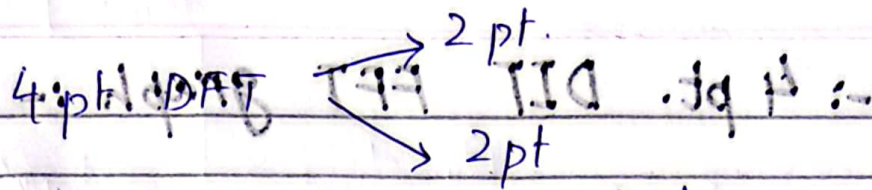
$N/4$ pt

$N/4$ pt

$N/4$ pt

$N/4$ pt

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It means N should be multiple of '2' if we want to split it into 2 branches

i.e. $N = 2^x$
 $\downarrow$
 Base Radix

Usually base of radix is prime number
 2, 3, 5, 7, ...

Radix - 2

DIT

DIF

Decimation in time
 (If we divide $x(n)$)

Decimation in freq.
 (If we divide $X(k)$)

∴ 4 pt. DIT FFT graph :-

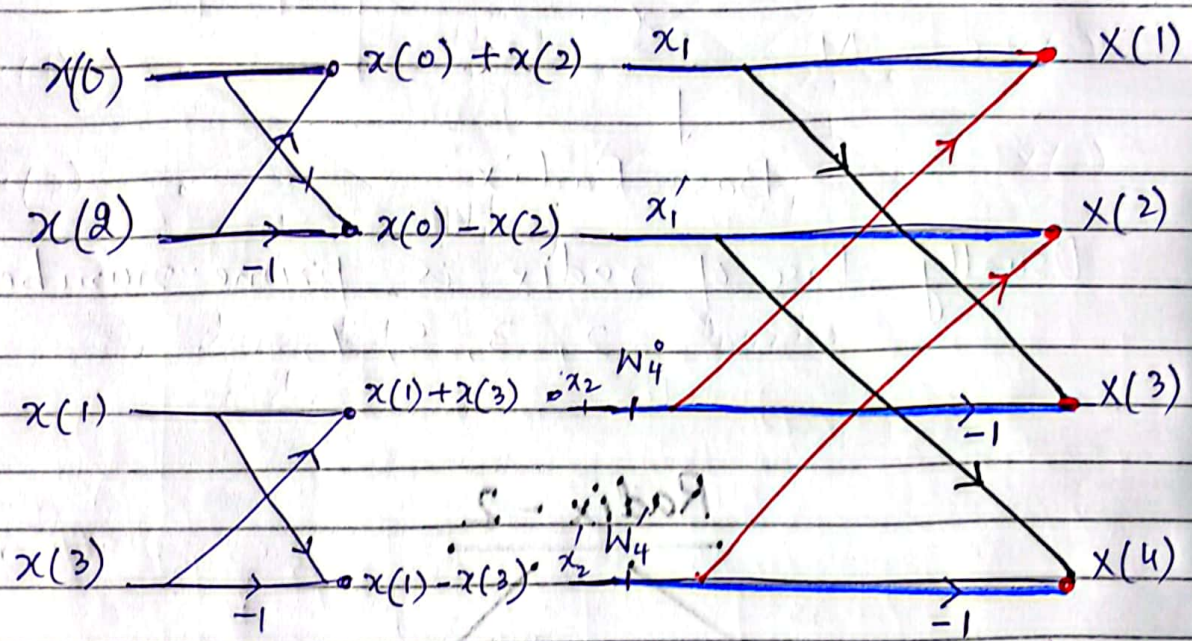
Dt means

$$x(n) = \{x(0), x(1), x(2), x(3)\}$$

(4 such sequences)

Combine $x(0) \text{ \& } x(2) \rightarrow$ even part

$x(1) \text{ \& } x(3) \rightarrow$ odd part



2nd stage

(i) 1st with 3rd value

2nd with 4th

(ii) 'x' last two branches with '2' values.

a) multiply with -1

b) multiply with W_N^k
i.e. $W_4^0 \text{ \& } W_4^1$

(11)

Suppose

$$x(0) + x(2) = x_1$$

$$x(0) - x(2) = x_1'$$

$$x(1) + x(3) = x_2$$

$$x(1) - x(3) = x_2'$$

$$\Rightarrow X(1) = x_1 + x_2 W_4^0$$

$$X(1) = x_1 + x_2$$

$$\Rightarrow X(2) = x_1' + x_2' W_4^1$$

$$= x_1' + x_2' (-j)$$

$$= x_1' - j x_2'$$

$$\Rightarrow X(3) = x_1 + (-1) x_2 W_4^0$$

$$= x_1 - x_2$$

$$\Rightarrow X(4) = x_1' + (-1) x_2' W_4^1$$

$$= x_1' - x_2' (-j)$$

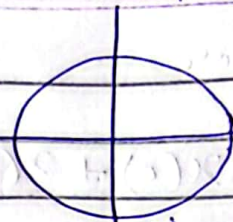
$$= x_1' + j x_2'$$

(11)

$$W_4^3 = j$$

(12)

$$W_4^2 = -1$$



$$W_4^0 = 1$$

$$W_4^1 = -j$$

 $x(n)$
 $x(k)$
 $x(0)$
 $x(0)$
 $x(2)$
 $x(1)$
 $x(1)$
 $x(2)$
 $x(3)$
 $x(3)$

$$W_4^0 = 1$$

$$W_4^1 = -j$$

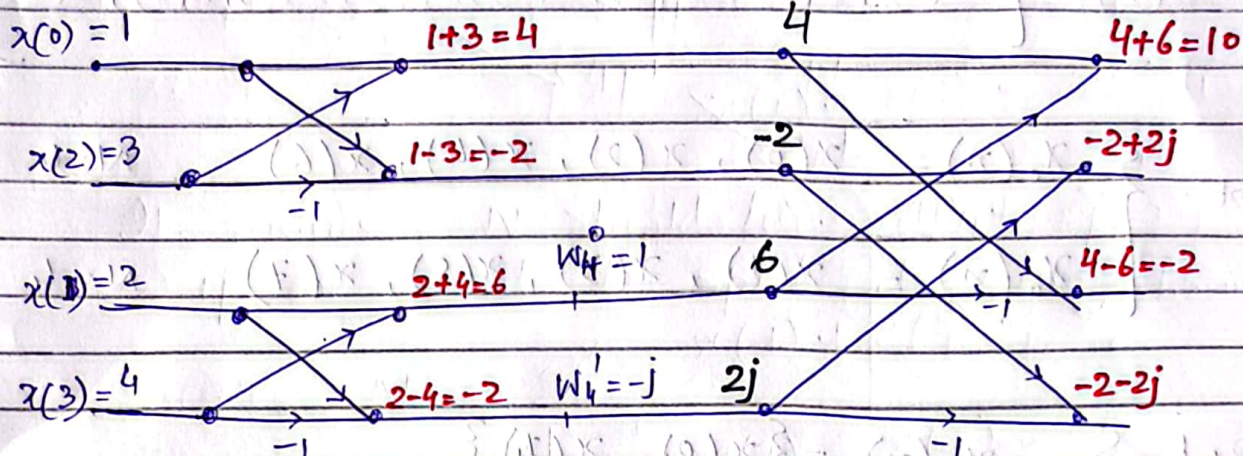
In decimated
order.

In order

(11)

(13)

Problem:- $x(n) = \{1, 2, 3, 4\}$



$$\therefore X(k) = \{10, -2+2j, -2, -2-2j\}$$